



ECONOMIC OPTIMISATION OF A FLOTATION PLANT THROUGH GRINDING CIRCUIT TUNING

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(Received 16 December 1999; accepted 9 June 2000)

ABSTRACT

The paper describes a procedure to tune a grinding circuit in order to maximise the economic efficiency of the flotation plant. The optimisation process is performed using a plant simulator based on phenomenological models of grinding and flotation and an empirical model for predicting the minerals size distribution from the ore size distribution of the grinding circuit product. The economic efficiency of the plant is assessed either by the net smelter return per ton of ore or the net revenue of the concentrator. The approach is illustrated for a lead-silver-gold plant located in Mexico and shows that the economic efficiency of this plant could be increased by 10 to 20%. © 2000 Published by Elsevier Science Ltd. All rights reserved.

Keywords

Flotation; grinding; process optimisation; liberation; simulation

INTRODUCTION

The overall objective of a grinding and flotation unit is to prepare a concentrate that maximises an economic performance index, which may be as simple as the net revenue of the plant. Practically, the links between the grinding and flotation circuits tuning and the economic objective are not obvious and the objectives are always broken down into particle size reduction, mineral liberation and mineral separation objectives (Herbst and Lo 1992, Lo *et al.* 1996, Herbst and Lo 1996). However, the optimisation of each stage of a chain of processes does not necessarily lead to the optimal operation of the overall chain, since the procedure neglects the interactions between the various units. Furthermore, it is generally difficult to define an optimisation criterion for each process unit well correlated to the overall economic objective. Usually, grinding circuit tuning requires a compromise between the throughput and the product fineness. The tuning of the flotation process results in a compromise between the concentrate grade and the recovery of valuable minerals.

The objective of this study is to determine an optimal grinding circuit tuning for the maximisation of the economic value of the concentrate produced by the subsequent flotation process. The complexity of defining optimisation criteria and optimum operating conditions has been extensively discussed in Kelly

and Spottiswood (1982) and Kelly (1991). Bascur (1991) has proposed a profitability concept defined by relating profit, recovery, tonnage and grind in order to co-ordinate the overall grinding and flotation objectives. He showed that there is an optimal profit as a function of the milling rates and proposed control strategies for maximising the revenue. Morrison (1993) discussed also various optimisation strategies such as minimum energy and maximum recovery philosophies and studied the impact of the grinding circuit tuning on the metal recovery. In those studies the right questions about optimisation of mineral processing plants are raised, but there is no complete methodology offered to optimise plant performance.

For an existing plant, one possible way to solve the performance optimisation problem is to try various set-ups of the grinding circuit and evaluate their effects on the flotation products. However, the true effects are usually of small magnitude and most of the time hidden by the numerous disturbances, which occur in a mineral processing operation. Plant simulators provide a noise free environment which allows possible answers to the problem of overall plant optimisation (Weller *et al.* 1996, Morrell *et al.* 1993, Apling and Montaldo 1982). Unfortunately, it is extremely difficult to link grinding and flotation simulators, because this operation requires a mineral liberation model to predict the particle composition as a function of their size. There are very good attempts to solve this problem theoretically, but they lead to complex models difficult to handle in simulation studies (Barbery 1991, Choi *et al.* 1988, Herbst *et al.* 1988, King 1979, Mehta *et al.* 1990, Wiegel 1976). In this paper, an empirical model is used to link the ore PSD (particle size distribution) to the PSDs of the various minerals of the grinding circuit product (Bazin *et al.*, 1994). Thus, in the flotation simulator, particles are characterised by the size and the nature of the mineral they are made of, while in the grinding circuit they are characterised only by their size. The key linking element is the conversion of the ore PSD into minerals PSDs.

The paper first presents the definitions of the metallurgical and economic performance criteria that are used to optimise a grinding/flotation unit. Then, the simulated plant, a silver-lead mineral processing operation, is presented together with a summary of the grinding, liberation and flotation models used for simulation. In a third part, the optimisation procedure is explained. It is divided into five steps, the results of which are presented in the last sections.

PERFORMANCE CRITERIA

Any mineral processing operation is viewed (Figure 1) as a plant consisting of one feed and two products (the tails and the concentrate). The multi-concentrate plants are not studied here, however the economic optimisation of these concentrators has been discussed in a previous paper (Bazin and Hodouin, 1997) emphasising the metal prices. Figure 1 divides the operation into two stages only, comminution and separation, although regrinding is a usual practice in the separation stage. The economic optimal configuration of comminution and separation units in the plant flowsheet is not discussed here, albeit it is an interesting topic that could be also studied by simulation.

The variables F , T and C (Figure 1) are respectively the ore feed rate, tails and concentrate flow rates, and f_i , t_i and c_i are the mass fractions of the n critical metals contained in the ore, the index 0 standing for the water. The solids recovery is $\rho = C/F$ and metal i recovery is $\rho_i = Cc_i/Ff_i$.

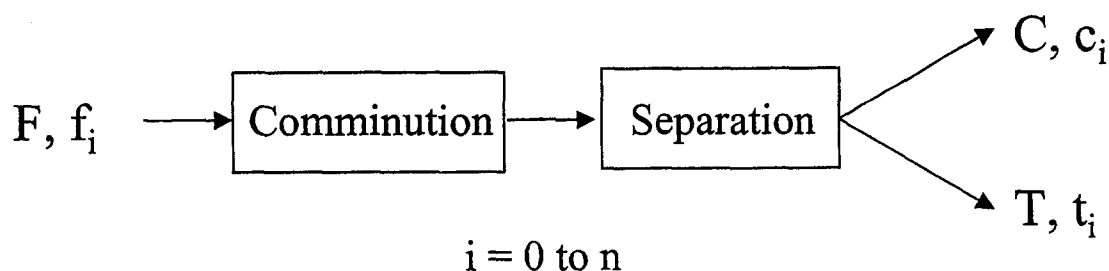


Fig.1 A mineral processing operation.

Metallurgical performance indicators are usually located in the grade–recovery plane (c_i, ρ_i) for the valuable metal and in the selectivity plane (ρ_j, ρ_i) where j is a metal to be rejected and i the metal to be recovered. A global economic performance index of the plant could be defined as the net revenue per unit of time R , given by the following equation:

$$R = FR_o - C_{Mn} - C_{Ml} \quad (1)$$

where R_o is the net smelter return (NSR) per ton of ore, C_{Mn} the mining costs per unit of time and C_{Ml} the milling costs per unit of time. The mining and milling costs can be divided into two parts, one (C_F) which corresponds to the fixed costs (salaries, capital...) and another one (C_V) which varies with the operating conditions:

$$C_{Mn} = C_{FMn} + C_{VMn} \quad (2)$$

$$C_{Ml} = C_{FMI} + C_{VMI} \quad (3)$$

C_{VMn} should normally vary with F , as well as C_{VMI} which depends on energy and reagents costs and on varying costs of ancillary equipment (pumping, dewatering, tails handling...).

The net smelter return per ton of ore is expressed as a function of the net smelter return per ton of concentrate R_c :

$$R_o = R_c \rho \quad (4)$$

where R_c is given by:

$$R_c = \sum_i v_i - C_T - C_S - \sum_i C_{Ri} - \sum_i P_i \quad (5)$$

In Equation (5), v_i is the payment for metal i per ton of concentrate (i obviously corresponds to payable metals). This variable is directly related to the free metal market prices and to the concentrate grades c_i through parameters defined by the mine/smelter contract. Usually it is a linear function of the metal content c_i . In Equation (5), C_T and C_S are the concentrate transportation and smelting costs per ton of concentrate. It can be either constant or a function of the metal market price and energy costs depending upon the sale contract. C_{Ri} is the refinery cost associated to the refining of metal i . It is usually proportional to the mass of metal, a variable that is related to the concentrate grade c_i . The variables P_i are penalties associated to impurities and water in the concentrate.

Assuming that mining and milling costs per unit of time are constant, the maximum net revenue is obtained through the maximisation of FR_o or CR_c . The dominant term in CR_c , assuming constant transportation and smelting costs per ton of concentrate and inactive penalties, is $\sum_i v_i = \sum_i \alpha_i C c_i$ for the valuable metals,

where α_i is the price paid for metal i . At constant plant capacity (F) and feed grades (a_j), $C c_i$ is maximised by increasing metal recoveries while maintaining grade at a value which does not penalise the NSR. At varying capacity, the maximum revenue can be obtained by pushing capacity while accepting grade and/or recovery decrease. There should however be a compromise between economic performances and metallurgical performances, since maximising the net revenue tends to decrease metallurgical performances and increase the losses of valuable minerals in the plant tailings. Low recovery strategies have ethical implications related to resource conservation and environmental concerns (Steane 1976, Kelly 1991) which are difficult to translate into penalties within an economic index.

In order to put emphasis on the metallurgical aspects of a mineral processing operation performance, one can define operation efficiency indices, i.e. criteria which assess the ability of the plant to exploit the potential metal value contained in the raw ore. For that purpose it is assumed that the ore is ideally processed, i.e. that the concentrate contains only the payable minerals of the plant feed. The concentrate quality can be evaluated by the following ratio:

$$E_q = \frac{R_c}{R_c^*} \quad (6)$$

where R_c^* is the net smelter return associated to the ideal concentrate, calculated by the application of the smelter contract clauses. Alternatively, the plant metallurgical performance could be assessed with respect to the plant feed rather than to the concentrate using the index defined by:

$$E_q' = \frac{R_o}{R_o^*} = \frac{R_c}{R_c^*} \frac{\rho}{\rho^*} \quad (7)$$

where ρ^* is the solids recovery for the ideal concentrate. The indices E_q or E_q' can be normalised by nominal values of R_c and R_o rather than by ideal values of R_c and R_o . They both characterise the success of the mineral liberation and separation processes, but do not inform about the plant capacity. In order to introduce both qualitative and quantitative aspects of the technical performance of the plant, one can define E_{qq} as:

$$E_{qq} = FR_o / (FR_o)_{ref} \quad (8)$$

where $(FR_o)_{ref}$ is a reference value for the evaluation of the plant performance. It can be selected as $F_{max}R_o^*$ where F_{max} is the maximum allowed capacity of the plant for the available equipment on the site. A nominal value $F^0R_o^0$, corresponding to the average behaviour of the plant, can also be used as a reference value.

An advantage of E_q and E_{qq} over R is that they are dimensionless numbers, which characterise the value of the plant output with respect to the potential value of the feed. The index E_{qq} is also an economic optimisation criterion when milling and mining costs have constant values. This is usually a reasonable assumption around the nominal operating point of the plant. Finally, it should be mentioned that the maximisation of the metallurgical performance index E_q is equivalent to the maximisation of an economic optimisation index E_{qq} when the plant throughput is constant.

THE CIRCUIT AND ITS SIMULATOR

The optimisation technique was applied to the grinding and flotation circuits of the Fresnillo Concentrator, a property of the industrial group Peñoles located in northern Mexico. The plant processes 2200 tons per day of a lead–silver ore. The main minerals present in the non-sulphide gangue (86.35%) are: gold (2g/t), pyrrargyrite (0.16%), galena (0.73%), arsenopyrite (0.85%), sphalerite (2.01%) and pyrite (9.90%) (Sosa-Blanco *et al.* 1999). Silver assays 950g/t and is mainly contained in pyrrargyrite.

The grinding circuit consists of a ball mill followed by an hydrocyclone classifier (Figure 2). There are two flotation circuits, which separate gold, silver and lead from sphalerite, pyrite, arsenopyrite and non-sulphide gangue. The first circuit processes the mine high-silver feed. The high-grade circuit tailings are classified and the fines treated by a low-silver circuit, while the coarse fraction goes to regrinding and is returned to the flotation feed. In this study, only the grinding and high-grade flotation circuits are optimised by simulation.

The grinding circuit is simulated using the traditional model involving breakage distribution and rate functions and perfect mixers for transportation through the mill. The Plitt's model is used for the hydrocyclone. For a complex sulphide ore, it has been observed (Bazin *et al.*, 1994, Edwards and Vien, 1999) that there are constant relationships between the ore cumulative PSD and the mineral cumulative PSDs for a given grinding/classification flowsheet. The difference in the behaviour of the various minerals is attributed to the selectivity of grinding and classification with respect to particle mineralogy and density. For each mineral a specific third order polynomial is used to predict the mineral PSD from the ore PSD. Figure 3 shows the polynomials calibrated to the data collected in the Fresnillo plant (Sosa-Blanco *et al.* 1999).

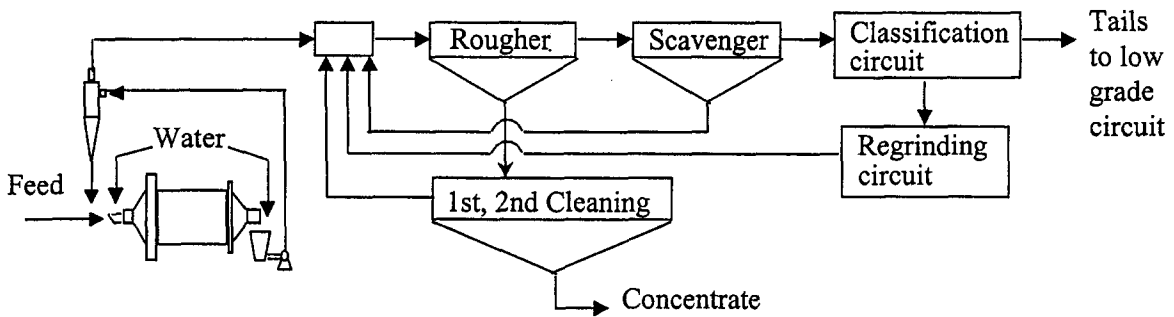


Fig.2 Grinding and high-grade flotation circuits.

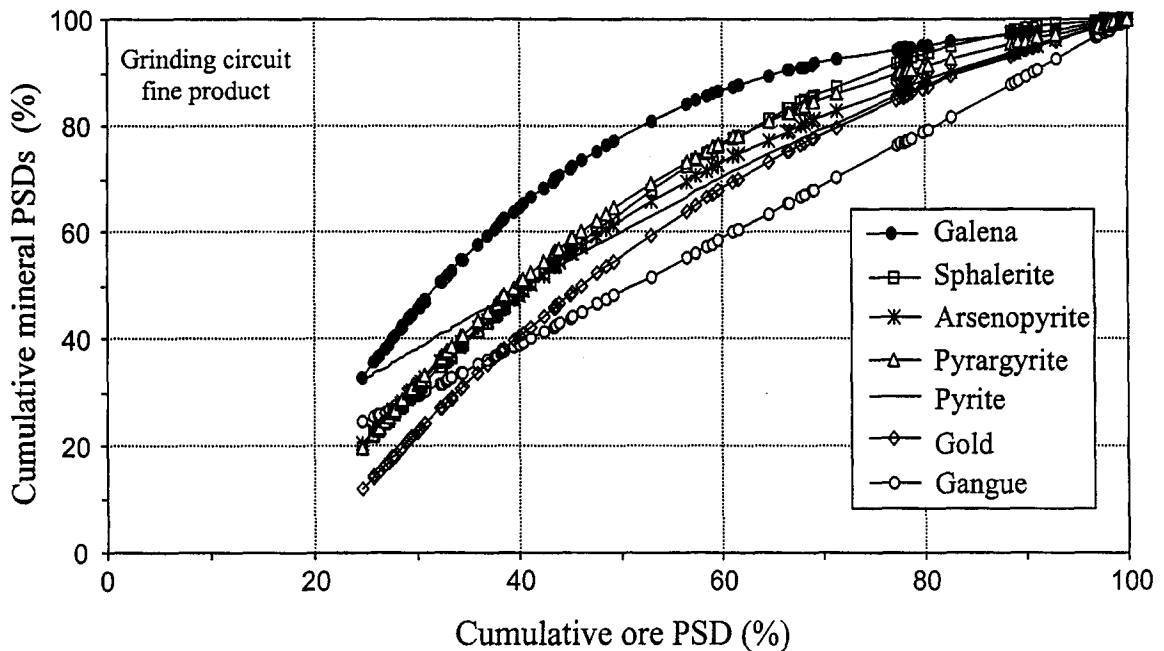


Fig.3 Relationship between ore and mineral cumulative PSDs.

The flotation model used for the simulation of the high-grade circuit is based on the following assumptions:

- the minerals in the flotation circuit behave as if they were independent particles having mineral- and size-specific flotation rates and entrainment factors. The requirement behind this assumption, as well as behind the above PSD relationships, is not that grinding produces only mono-mineral particles, but that the particle composition distributions are reasonably invariant in the various size classes. This assumption is not as restrictive as ignoring the effect of locked particles, an unlikely situation; however it requires that the liberation degree is constant in each particle size interval. The variation of the overall degree of liberation is thus only related to the variation of the PSD. As a consequence of this assumption, the flotation rates as well as the entrainment factors of the various minerals must be understood as resulting of a weighted average of the behaviour of the particles in a given size interval as a function of their degree of liberation.
- the slurry in the flotation cells is perfectly mixed.
- the cleaning stages exhibit constant separation efficiencies, i.e it is assumed that a constant fraction of each particle class is directed to the cell concentrate.
- the hydrocyclone in the regrinding circuit has constant classifier efficiency.

- the regrinding circuit exhibits the same breakage distribution and the same relationship between ore and minerals PSDs as the primary ball mill circuit.

The simulator calibration and validation has been detailed in Sosa-Blanco *et al.* (1999). Considering the complexity of the plant to be simulated and the number of minerals involved in the process, the simulator adequately reproduces the circuit metal recoveries and stream compositions.

THE OPTIMISATION PROCEDURE

To study the optimal tuning of the Fresnillo plant, it is assumed that C_{Mn} and C_{Ml} contain only fixed costs. The reagent cost variations as well as the grinding media and energy consumption variations are therefore neglected in the operating domain explored by the simulation.

The net smelter return R_c is calculated using the confidential equations of v_{Ato} , v_{AG} , v_{Pb} , C_T , C_S , P_{As} , P_{Zn} , C_{RAto} and C_{RAG} described in the Peñoles smelter contract. The simulation results and metal prices are incorporated into these equations to evaluate the performance of the overall grinding/flotation unit with the following three criteria:

$$E_1 = R_c/R_o^* \quad (9)$$

$$E_2 = R_c/R_o^0 \quad (10)$$

$$E_3 = FR_o/F^0 R_o^0 \quad (11)$$

The indices E_1 and E_2 are of the E_q type, while E_3 is of the E_{qq} type. The variables E_1 and E_2 are used to assess metallurgical performances, while E_3 is an economic performance indicator.

As the flotation simulator is not able to predict the effects of the reagent addition variations, it is assumed, in the search for an optimum operating regime of the grinding/flotation unit, that the model parameters (rates of flotation and entrainment factors for the rougher-scavenger stage, and recoveries for the cleaner stages) are invariant in each class of particles described by their size and mineral type. The flotation circuit design variables are also maintained at constant values, and then the only sources of flotation performance variations are the mineral PSDs and the ore and water flow rates of the simulated product of the grinding circuit.

The input variables of the grinding circuit which are involved in the optimisation procedure are design variables such as the hydrocyclone arrangement (one-stage or two-stage classification) and geometry (diameter D , height H , and inlet diameter IN), and tuning variables such as the two water addition rates (W_1 at the mill discharge and W_2 at the mill feed), the fresh ore feed rate (F) and the hydrocyclone apertures (vortex finder VF and apex diameter AP). Other design or tuning variables such as the mill diameter, the mill RPM, the grinding media load and size distribution are left outside the optimisation algorithm and assumed to be constant. The optimisation problem is thus to maximise either E_1 , E_2 or E_3 with respect to F , W_1 , W_2 , VF , AP , D , H and IN for different arrangements of the classifier (one-stage or two-stage). Theoretically, one could vary all these parameters, simulate the grinding/flotation unit, calculate the economic performance indices and change the parameters according to a strategy leading to the maximum value of E_1 , E_2 or E_3 . The drawback of such an approach is that it would be extremely difficult to trace and understand the role of each independent optimisation factor as well as their synergetic effects. A strategy based on a step by step approach where a limited number of variables is considered at each step is preferred. It allows a deeper understanding of the critical factors in the search for an optimum, and progressively orients the search for optimal operating conditions. The following optimisation procedure was followed:

Step 1: the impact of the ore PSD on the flotation performance is studied. For that purpose artificial ore PSDs are generated independently of the behaviour of the grinding circuit, which is not simulated at this stage of the procedure;

Step 2: since the ore PSD is strongly related to the classifier efficiency, hydrocyclone separation curves are artificially generated by manipulating the corrected cut size (d_{50c}), the by-pass coefficient (R_p) and the classifier sharpness (m), i.e. the slope of the corrected efficiency around the cut size d_{50c} . At this stage of the procedure both grinding and flotation circuits are simulated;

Step 3: when the role of the classifier efficiency is understood, it is then possible to manipulate the tuning or design variables W_1 , W_2 , AP , VF , D , IN and H at constant circuit capacity F to search for a classifier efficiency and grinding circuit behaviour which maximise the performance indicators previously defined;

Step 4: from step 3 the role of the grinding circuit is clarified. This allows at step 4 to perform the optimal tuning of the most significant variables W_1 , W_2 , VF and AP while varying F and taking into account operating constraints, such as mill and cyclone % solids and pumping capacity, which are requested for a realistic operation of the grinding circuit.

Step 5: once the optimal tuning is performed for a one-stage classification circuit, it is finally possible to assess the performances of two-stage classification grinding circuits.

The results of these five steps of the optimisation procedure are now presented and discussed.

RESULTS OF OPTIMISATION STEP 1

The objective of this first step was to characterise the impact of the flotation feed PSD and % solids on the flotation performance at constant operating conditions of the flotation circuit (fixed rates of flotation, entrainment factors, regrinding circuit tuning, and recoveries in the cleaning cells). Six values of the feed % solids were used between 32 and 41% (32, 33.7, 35.4, 37, 39, 41). The PSDs were described by the following relationship:

$$y = c + (100 - c) [1 - \exp(-(x/a)^b)] \quad (12)$$

where y is the percentage of particles smaller than x , and a , b , c are coefficients. The artificial PSDs are shown in Figure 4. The performance index E_1 (Equation 9) is presented in Figure 5 for the thirty simulated flotation operating conditions. First, it is clear that an increase of the slurry % solids improves the plant efficiency. This is simply related to the increase of the mean residence time (MRT) in the flotation unit, a factor which enhances the valuable metals (Au, Ag, Pb) recoveries, while decreasing the concentrate grades (Figure 6). The plant efficiency is improved because the positive impact of the recovery increase is larger than the negative impact of the grade decrease.

With respect to PSD, it is clear that PSDs no. 1 and 4 give better results than the nominal PSD (no.2) and that PSDs 3 and 5 strongly deteriorate the economic efficiency. Figures 4 and 7 show that it is very important to eliminate coarse particles, as in PSDs 1 and 4, in order to improve efficiency. The comparison of the merits of PSDs 1 and 4 in the fine size range is less evident since PSD1 contains 38% of $-10 \mu\text{m}$ and PSD4 only 16% of $-10 \mu\text{m}$, for almost the same performances. Looking at Figure 6, PSD 4 seems to be potentially better since it gives operating conditions closer to the 100% grade/100% recovery corner of the grade-recovery plot. However at constant flotation feed % solids PSD 1 gives a better economic efficiency since the decrease of grade is compensated by a higher silver recovery.

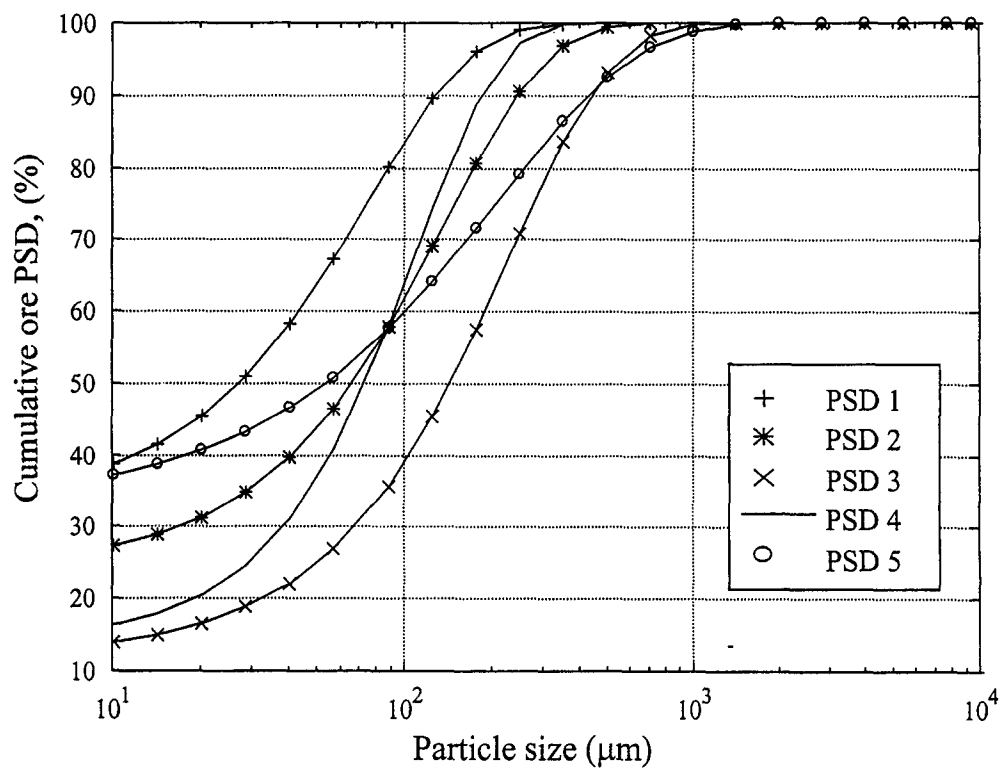
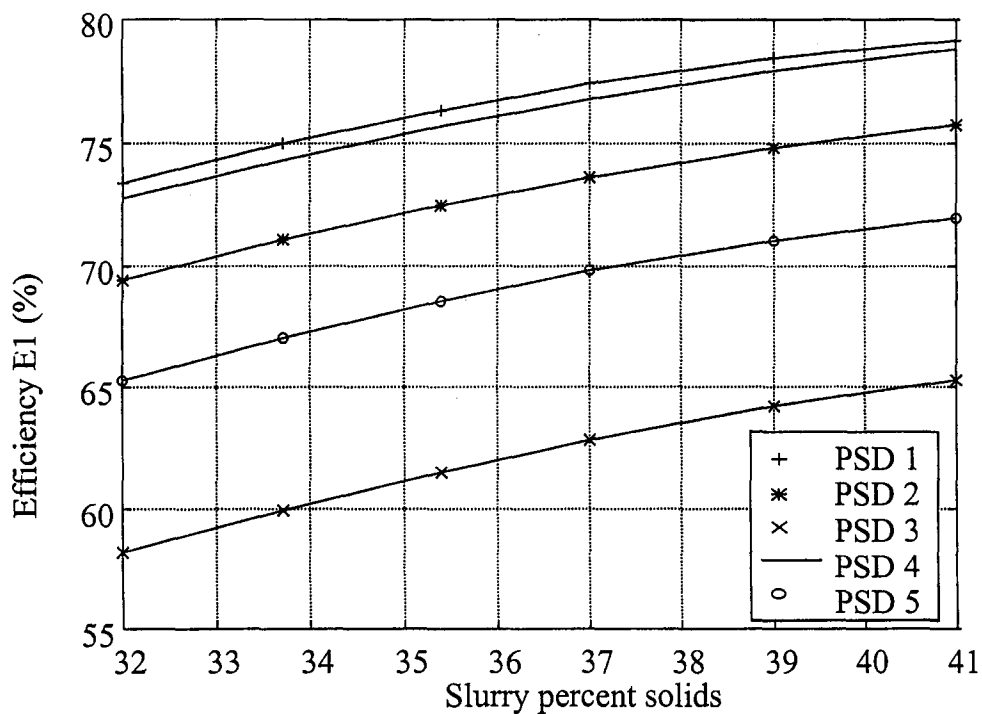


Fig.4 Synthetic PSDs of the fine product.

Fig.5 Efficiency E₁ as a function of flotation feed % solids and PSD.

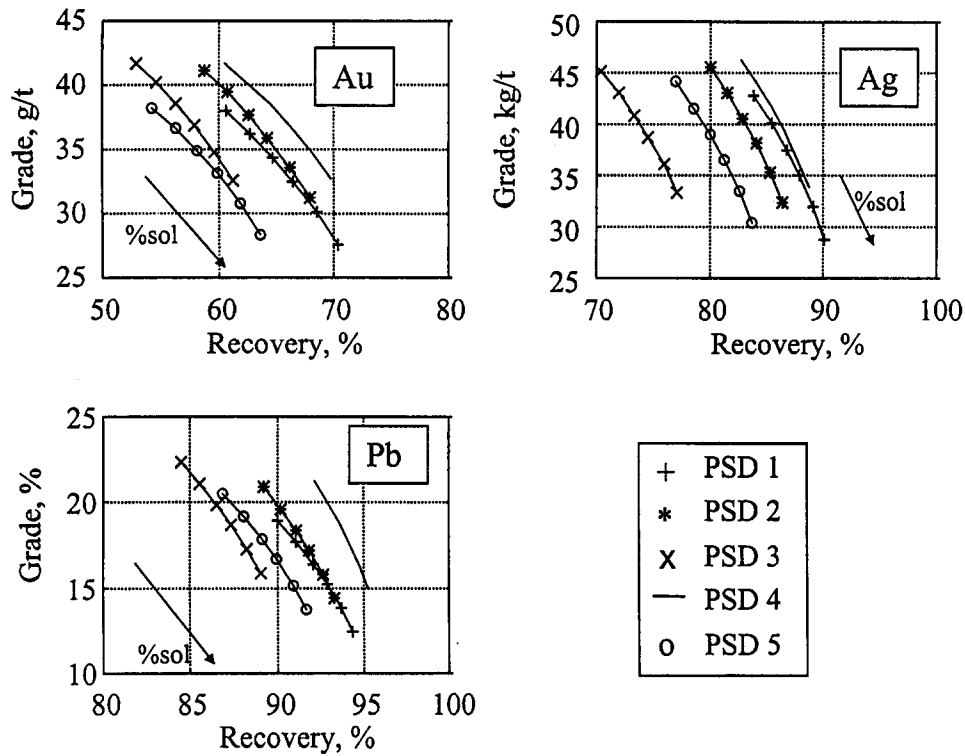


Fig.6 Grade-recovery plots for Au, Ag and Pb as functions of flotation feed % solids and PSD.

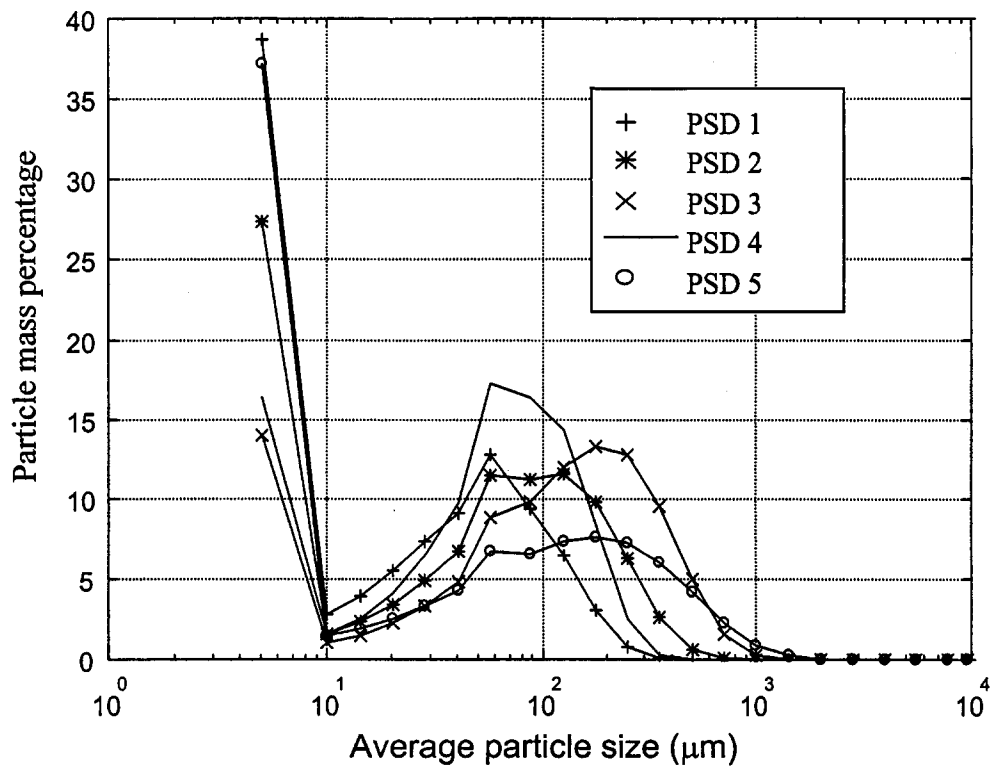


Fig.7 Size distributions of the synthetic grinding circuit products.

Since pyrrargyrite is the most valuable mineral of this ore, its behaviour in the flotation circuit must be carefully analysed. Figure 8 shows the pyrrargyrite PSD as well as its flotation rate constant in the

rougher-scavenger, recovery in the cleaners (independent on the operating conditions) and recovery in the rougher-scavenger (for the nominal operating conditions and PSD no. 2) as a function of the pyrrargyrite particle size. One can see that the maximum efficiency is roughly in the range 5–50 μm , thus explaining the good performances of PSDs 1 and 4. Figure 9 shows another reason why PSDs 1 and 4 exhibit better economic efficiency: they produce a better selectivity of the flotation circuit for pyrrargyrite against sphalerite, pyrite, arsenopyrite and non-sulphide gangue.

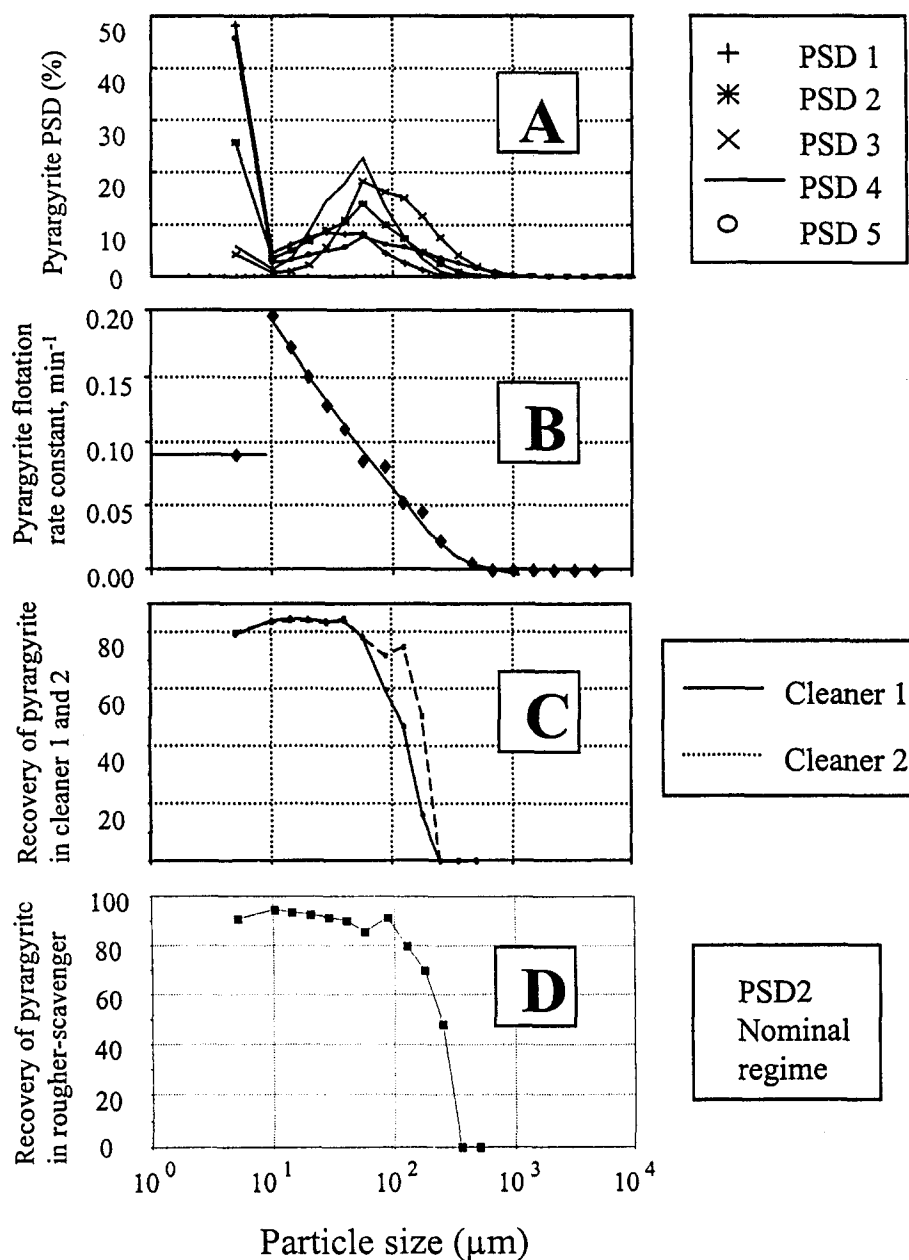


Fig.8 Properties of pyrrargyrite as a function of particle size (A: PSDs; B: rate constants; C: cleaner recovery; D: recovery in the rougher-scavenger at nominal operating conditions).

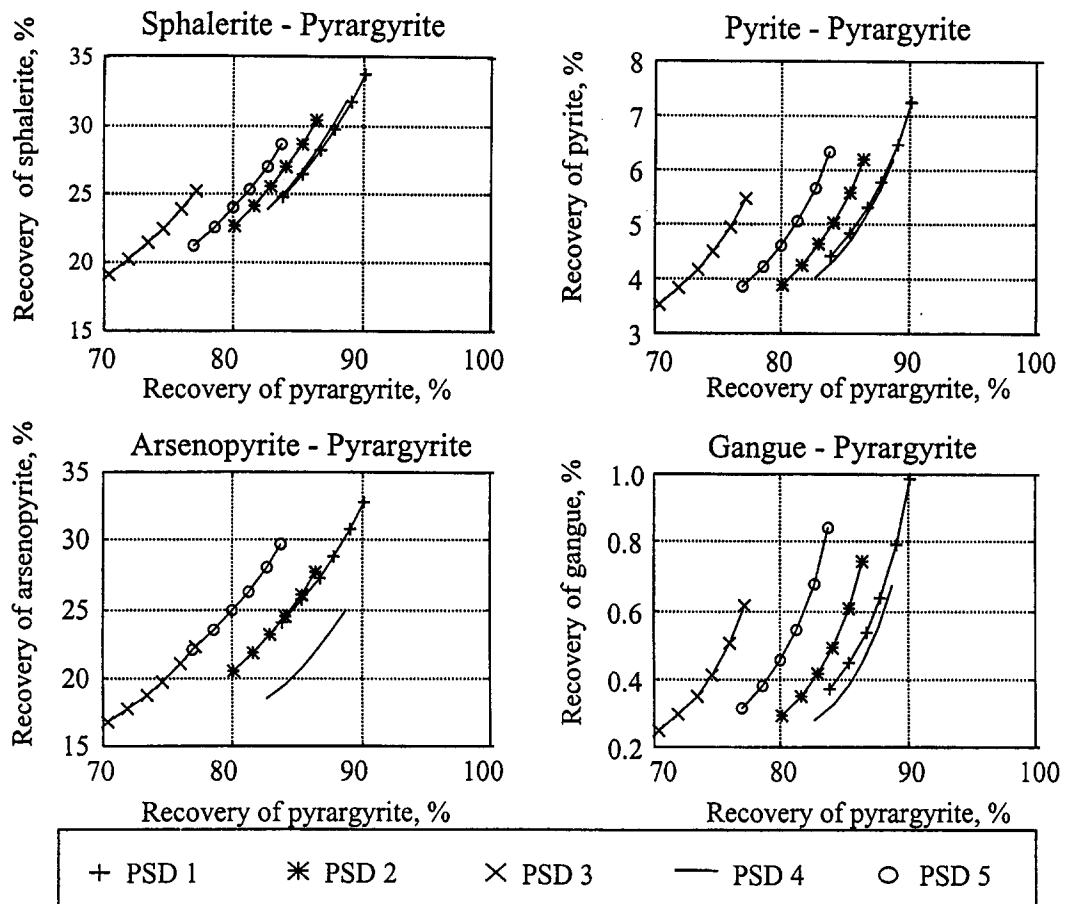


Fig.9 Selectivity of pyrrargyrite flotation against sphalerite, pyrite, arsenopyrite and gangue.

Assuming that PSDs 1 and 4 could be obtained by grinding, PSD4 should probably be preferred since it is globally coarser and thus requires less grinding energy. However the production of a PSD of type 4 would require a better size classification stage than producing a PSD of type 1, since its slope is steeper.

RESULTS OF OPTIMISATION STEP 2

The objective of Step 2 is to tune the classification process of the grinding circuit, in order to produce PSDs close to PSDs 1 or 4 of Step 1, while assuming constant rates of breakage and fragment distribution in the ball mill. The fresh ore feed rate is maintained at a constant value (98 t/h) as well as the ore composition. The actual efficiency of an hydrocyclone can usually be entirely represented by a reduced efficiency curve, parameterised by its sharpness m , and by the values of the corrected cut size d_{50c} and the by-pass coefficient R_f (Flintoff *et al.*, 1987). Therefore, these three parameters are varied according to a three-level experimental design (Table 1) generating nine simulation tests. Figures 10 and 11 show respectively the classifier efficiency curves and the corresponding PSDs produced by the simulator.

The economic efficiency E_1 is improved when d_{50c} and R_f are small and when m is large. This is logical according to the results of optimisation step 1 which favours finer PSDs with steeper slopes. The best combination is obtained with the simulation test 8. The economic efficiency (77.2%) is even better than those obtained with PSD1 and PSD4 in the previous step (76.7 and 76.0% respectively).

TABLE 1 Simulation results for various classifier efficiencies (fresh ore feed rate: 98t/h)

Test	Classifier efficiency			Circulating		
	d_{50c} (μm)	m	R_f (%)	% - 74 μm	load (t/h)	E_1 (%)
1	165	1.44	23	50.9	101	71.8
2	100	1.44	23	56.8	152	75.0
3	230	1.44	23	47.8	84	69.8
4	165	0.12	23	46.5	89	69.3
5	165	2.31	23	52.9	110	73.0
6	165	1.44	12	51.9	83	72.4
7	165	1.44	35	50.1	127	71.3
8	100	2.31	12	61.4	156	77.2
9	230	0.12	35	43.7	98	67.5

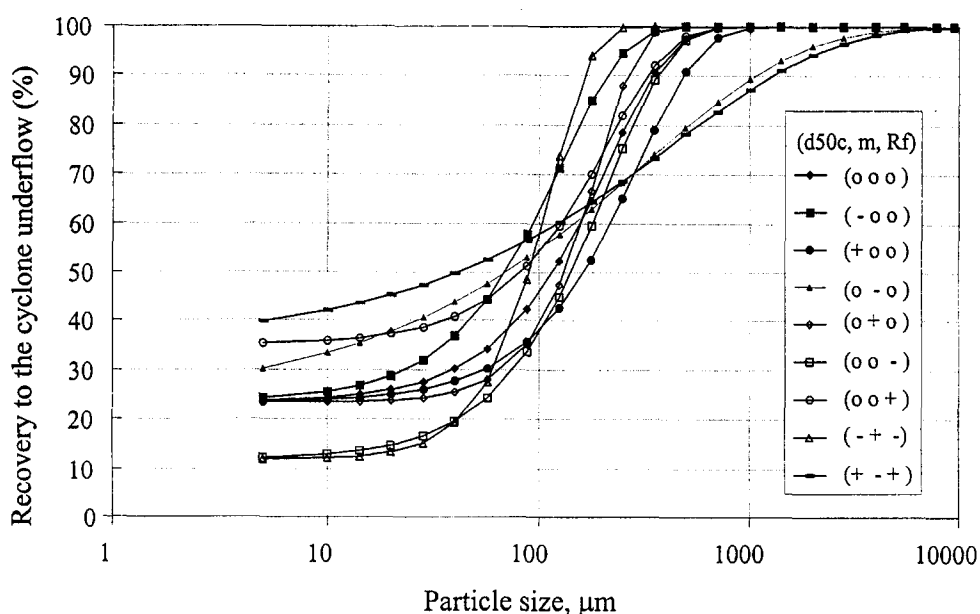


Fig.10 Classification efficiency curves for various R_f , d_{50c} and m values (the coded levels -, o, + correspond respectively to the low, medium and high values of Table 1).

The above nine tests were conducted at 36% solids for the grinding circuit product. However, as seen in the first optimisation step, the percentage of solids modifies the plant performance. It was then decided to change the % solids and to simulate, for the new operating conditions, the same tests as above. The results are presented in Figure 12. Clearly at constant % solids and feed rate F , the efficiency E_1 is directly a function of the particle fineness, expressed here as the % - 74 μm . The efficiency is monotonously increasing with the % - 74 μm , without exhibiting a maximum value. This is due to the fact that the average flotation rate continuously increases up to 10 μm and is still high for the - 10 μm particles. Undoubtedly, the rate of flotation decreases for fine particles, and extended PSD in the fine size range would have shown a decrease of the flotation efficiency. The effect of the % solids is caused here neither by the behaviour of the hydrocyclone, since the classifier efficiency is not related to the dilution, nor by the behaviour of the mill, since the slurry rheology is not taken into account in the breakage model. Its effect is related to the change

of the MRTs in the grinding mill and the flotation circuit. As a matter of fact, the latter explains almost all the observed phenomenon.

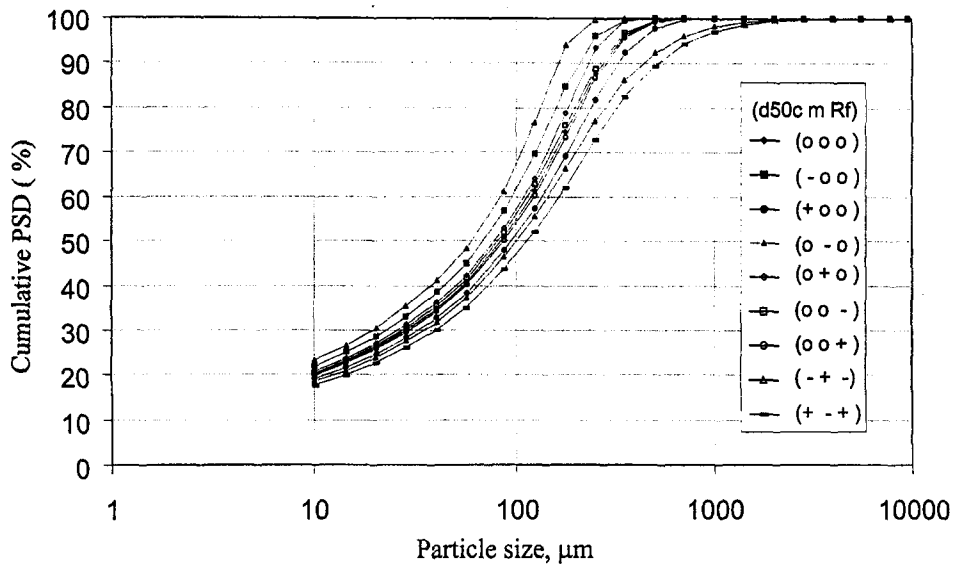


Fig. 11 PSDs produced by the classifier efficiencies of Figure 10.

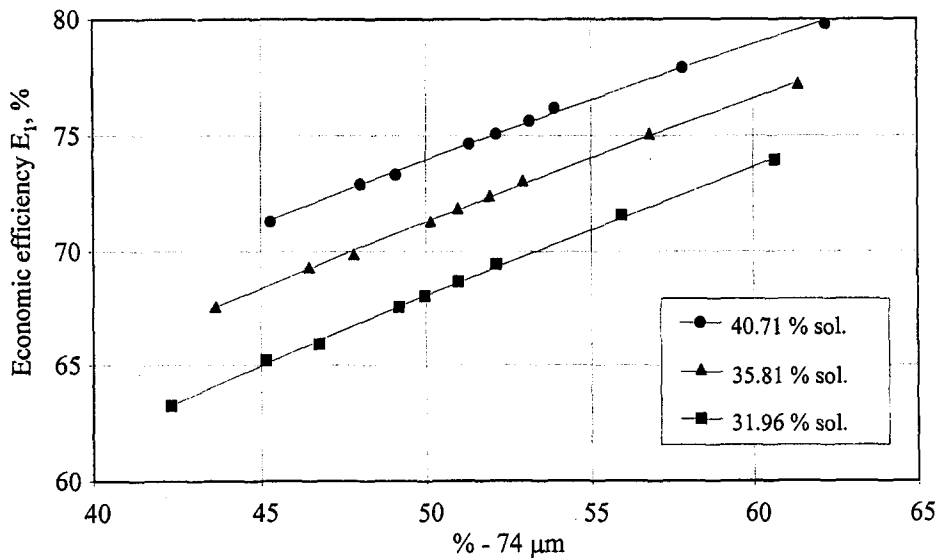


Fig.12 Economic efficiency as a function of flotation feed % solids and % ~74 μm .

RESULTS OF OPTIMISATION STEP 3

Optimisation step 2 showed that it would be possible to obtain better flotation feed PSDs, with respect to maximisation of economic efficiency at constant feed rate, by changing the classifier performance through the manipulation of d_{50c} , R_f and m . Also it showed that, around the normal range of operation of the grinding circuit, the economic efficiency at constant feed rate and % solids is strongly correlated to the flotation feed %~74 μm .

The objective of optimisation step 3 is to look for grinding operating conditions which improve the classifier efficiency in the direction pointed at step 2, while maintaining the ore feed rate at its nominal value ($F = 98$ t/h). For that purpose, the grinding circuit was simulated at constant feed rate for an experimental design where the two water addition rates W_1 and W_2 as well as the five hydrocyclone geometric factors are varied according to the levels defined in Table 2. The results of the sixteen simulation tests are given in Table 3.

TABLE 2 Levels of the grinding circuit tuning parameters

Hydrocyclone Parameter	Parameter values		
	-	0	+
Water addition rate W_1 (m ³ /h)	35.0	42.5	50.0
Water addition rate W_2 (m ³ /h)	13.0	17.0	21.0
Apex diameter (cm)	5.08	5.72	6.35
Vortex finder diameter (cm)	8.89	10.8	12.7
Inlet diameter (cm)	12.7	15.2	17.8
Cyclone diameter (cm)	42.8	50.8	58.8
Cyclone height (cm)	143	168	193

The results of Table 3 show that it is difficult to find tuning conditions that decrease d_{50c} and R_f , and increase m , for improving the %–74 μm , and, at the same time, increase the flotation feed % solids, for increasing the flotation circuit and grinding mill MRTs. Looking at the efficiency criteria E_i , the best performances are obtained by reducing water addition (tests 2, 4 and 16), due to the MRT increase in the flotation circuit. Test 16 gives better results because at the same time the %–74 μm is increased, due to the improvement of m and R_f , although d_{50c} is coarser.

Since the MRT variation in the flotation circuit tends to counterbalance the effects of the grinding circuit improvements which could be obtained by increasing water addition, the criteria E_f was replaced by E'_f , which assumes that the flotation feed % solids is at its nominal value whatever the water addition in the grinding circuit. This would require the introduction of a thickener between the grinding and flotation circuits, or adding cells in the flotation circuit. Tests 3, 6, 10 and 16 provide the best results confirming that maximisation of %–74 μm at constant flotation MRT is the way to improve the plant performance.

Another conclusion from optimisation step 3 is that some combinations of water addition rates and hydrocyclone geometry tunings produce operating conditions, such as hydrocyclone underflow roping, which could not be implemented in the actual grinding circuit, because of operating constraints violations. Finally the results confirm the usual assumption that the most important tuning variables for the hydrocyclone are the feed percent solids and the apex and vortex finder diameters.

TABLE 3 Simulation results as a function of the tuning parameters

Test	Tuning Parameter							Classifier efficiency Coefficients			% -74 μm	% sol	Circulating load t/h	E_1 %	E'_1 %
	W_1	W_2	AP	VF	IN	D	H	d50c μm	m	Rf %					
1	o	o	o	o	o	o	o	161	1.44	23	50.9	35.8	101	71.8	71.8
2	-	o	o	o	o	o	o	183	1.44	25	50.0	36.8	96	73.1	71.4
3	+	o	o	o	o	o	o	145	1.44	22	51.7	34.8	107	70.3	72.2
4	o	-	o	o	o	o	o	170	1.44	25	51.3	36.3	97	73.0	72.0
5	o	+	o	o	o	o	o	158	1.44	22	50.7	35.3	105	70.6	71.7
6	o	o	-	o	o	o	o	173	1.67	15	51.6	35.8	86	72.2	72.3
7	o	o	+	o	o	o	o	154	1.25	32	50.2	35.8	120	71.4	71.4
8	o	o	o	-	o	o	o	139	1.19	33	50.8	35.8	130	71.8	71.8
9	o	o	o	+	o	o	o	185	1.66	16	50.8	35.8	85	71.7	71.8
10	o	o	o	o	-	o	o	154	1.46	22	51.7	35.8	103	72.3	72.2
11	o	o	o	o	+	o	o	172	1.42	25	50.3	35.8	101	71.4	71.4
12	o	o	o	o	o	-	o	154	1.28	27	50.7	35.8	110	71.7	71.7
13	o	o	o	o	o	+	o	172	1.59	21	51.1	35.8	95	71.9	71.9
14	o	o	o	o	o	o	-	170	1.46	22	50.8	35.8	96	71.7	71.7
15	o	o	o	o	o	o	+	159	1.42	25	51.1	35.8	106	71.9	71.9
16	-	-	-	o	-	o	o	191	1.69	17	51.7	37.3	77	74.8	72.2

RESULTS OF OPTIMISATION STEP 4

The objective of step 4 is to optimise the grinding circuit with a one-stage classifier with respect to F , W_1 , W_2 , AP , and VF while satisfying operating constraints. From the tests of step 3, it became clear that constraints must be defined on some critical process variables. First, a maximum value of the volumetric percentage of solids at the cyclone underflow is defined to avoid roping conditions. It is calculated from a relationship proposed by Flintoff *et al.* (1987). The maximum value of the % solids in the ball mill is selected to be 80%, a value above which the grinding efficiency decreases (Laplante, 1983, Klimpel, 1982). The maximum mill throughput is selected as 205 t/h. The maximum water addition rate at the circuit sump box is chosen as 150 m³/h. These two last constraints are chosen according to the capacity of the equipment at the Fresnillo Mill.

As a first move toward the optimal operating regime, the optimal distribution of water between the mill and the hydrocyclone is looked for at constant fine product % solids (constant values of $W_1 + W_2$ and F) and constant hydrocyclone apertures. Table 4 shows that it is possible to increase the economic efficiency by 2% in comparison with the reference state ($E_2 = E_3 = 102\%$ in Table 4), while respecting all the constraints. From Table 4 one can see that the efficiency increase is due to the improvement of the flotation feed fineness (54.1% - 74 μm instead of 50.9% - 74 μm). At constant feed rate, E_2 and E_3 are identical and are both improved. When F is considered as an additional degree of freedom, it is possible to improve E_3 , while deteriorating E_2 , by an increase of F up to a limit corresponding to a constraint violation. In order to avoid a too drastic decrease of the metallurgical performances, measured by E_2 , one can also maximise E_3 while maintaining the % - 74 μm at its nominal value. The result is presented in Table 4. The production is increased from 98 t/h to 115 t/h for 11% increase of the revenue (measured by E_3). The metallurgical

performance index E_2 is decreased to 93%, clearly showing that a production increase may improve the net revenue, but at the cost of losing valuable minerals in the plant tails.

TABLE 4 Results of optimisation steps 4 and 5

Test	Classifier	Process Variables	% -75 μm	F (t/h)	Product % sol.	E2(%)	E3(%)	Grade Ag (Kg/t)	Silver Recovery
1		Reference	50,9	98	35.8	100	100	39.5	82,2
2		W_1, W_2	50,9	115	35.8	93	111	41.8	78,1
		$W_1+W_2=\text{cst}$	54,1	98	35.8	102		39.4	83,6
3		W_1, W_2, AP, V	50,9	123	27.5	90	116	42.7	76,3
			57,2	98	18.0	105		39.3	85,1
4		W_1, W_2, AP, V	50,9	133	13.4	87	121	43.7	74,3
			57,9	98	10.7	105		39.2	85,6
5		W_1, W_2, AP, V	50,9	131	24.3	88	120	43.4	74,8
			58,0	98	18.5	105		39.2	85,5
6		W_1, W_2, AP, V	50,9	113	44.4	94	111	41.6	78,4
			54,5	98	37.1	103		39.4	83,9
7		W_1, W_2, AP, V	50,9	121	43.2	91	115	42.5	78,8
			57,1	98	30.8	105		39.3	85,1

In the following searches for optimal tunings, the five process variables F , W_1 , W_2 , VF , and AP were selected, since the other geometric variables of the hydrocyclone do not have significant effects as shown by optimisation step 3, and, furthermore, cannot be easily changed in the plant. Since the flotation simulator would have been called a large number of times during the optimisation process, thus leading to large computation times, the algorithm of Figure 13, which does not require flotation simulation for the optimisation with respect to W_1 and W_2 , is preferred. Instead of optimising the economic efficiency, the % - 74 μm is maximised. As shown previously, at constant flotation feed % solids (here at the nominal value 35.8%), and constant ore feed rate F , the % - 74 μm is directly correlated to the economic efficiency. Since the flotation feed % solids is maintained at its nominal value, the method leads to a grinding circuit fine product % solids different from the flotation feed % solids. This is to enhance the effect of the grinding circuit tuning on the economic efficiency by avoiding too drastic changes of the MRT in the flotation circuit, as already mentioned above.

The variations of F , AP and VF are planned according to a grid of variations. As in the previous case, we looked at two different directions in the 3D space F , AP , VF : on the one hand, the circuit is optimised at constant capacity ($F = 98$ t/h), and on the other hand at constant product fineness (50.9% - 74 μm). As above, all the operating constraints are enforced. The first direction is interesting when the mining capacity is the bottleneck, while the second one is a reasonable constraint to maintain the metallurgical performances at acceptable levels. At nominal tonnage, an additional 3% of economic and metallurgical performances can be obtained by tuning AP and VF in addition to W_1 and W_2 ($E_2 = E_3 = 105\%$). When keeping the % - 74 μm at its nominal value, it is also possible to push the circuit capacity at 123 t/h for an increase of E_3 up to 111%. Despite the % - 74 μm being maintained at its nominal value, the metallurgical efficiency is decreased to 90 %. This is mainly due to the decrease of the MRT in the flotation circuit, because of the pulp flow rate increase by 63 t/h at constant feed % solids (35.8%). The economic efficiency E_3 could possibly be increased by accepting a coarser grinding circuit product, i.e. more silver losses in the high-grade circuit reject. The impact on the final plant tailings should also be considered, by taking into account the low-grade circuit. Most probably there will be more silver losses, a situation which might not be optimal looking at its long-term impact.

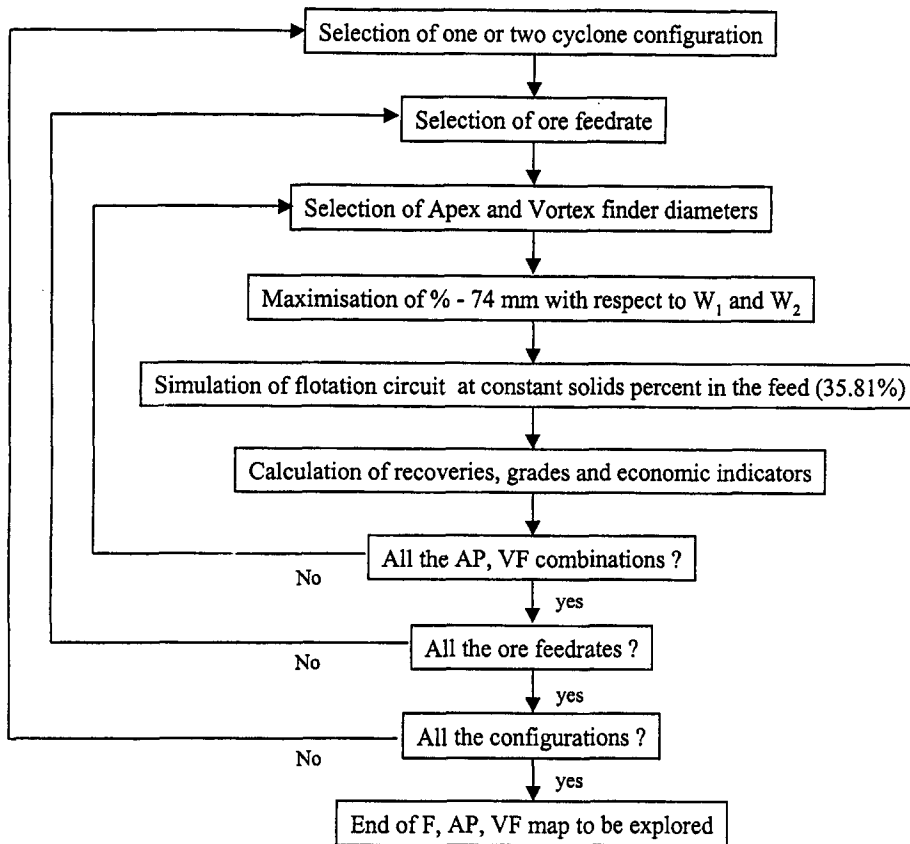


Fig.13 Optimisation procedure.

RESULTS OF OPTIMISATION STEP 5

Multiclassification is a way to improve the classification efficiency curve, and thus produce better grinding circuit products (Gelpe *et al.* 1985). Two options are available for two-stage classification systems: cleaning of the first stage overflow for producing a finer flotation feed (tests 4 and 5 of Table 4), or cleaning of the first stage underflow to decrease the by-pass coefficient R_f (tests 6 and 7 of Table 4). In both cases the second stage underflow (or overflow) may be recycled to the first-stage feed (tests 5 and 7) or not recycled (tests 4 and 6). Results of eight simulation runs (2 per test) are presented in Table 4. One simulation run is performed at constant production capacity (98t/h), and the other one at constant product fineness (50.9% - 74 μm). Reclassification of the underflow produces better results than overflow cleaning. Recycling of the second stage overflow does not significantly improve the circuit performance but asks for less water than the open classification circuit configuration. The economic efficiency is increased by 20% ($E_3 = 120\%$), while the metallurgical performance is decreased by 12% ($E_2 = 88\%$) at constant PSD. At constant tonnage the economic and metallurgical efficiencies are both improved by 5% ($E_2 = E_3 = 105\%$), i.e. at a level identical to the full optimisation of the one-stage classifier.

CONCLUSION

The study shows that it is possible to increase the silver recovery (as well as the gold and lead recoveries) by decreasing the coarse particle fraction (+74 μm). At constant production, this leads to a 5% increase of the net revenue. For this specific ore, the flotation rate is still high for - 10 μm particles. Thus, it is not possible to observe the decrease of performance which would appear for very fine grinding. For an ore

exhibiting a coarser liberation size, the same method would probably point at an optimal choice of the average particle size, not located on the fine size constraint. For the studied ore, more detailed analysis in the fine size range would have probably revealed the presence of an unconstrained maximum efficiency as a function of grinding product fineness

In practice to obtain a finer and steeper particle size distribution of the grinding product, it is necessary to improve the classifier efficiency, i.e. to increase the separation curve sharpness, and decrease the cut-size and the by-pass coefficient. Because of the constraints created by the hydrocyclone behaviour it is difficult to control separately these three factors. Roughly, increasing the total water addition in the circuit and distributing the water in such a way that the slurry % solids at the cyclone feed is minimised, improves the performance of the classifier. Using simultaneously smaller vortex finder and apex also improves the classifier efficiency. Alternatively, multi-stage classifiers can improve the classification efficiency and give more flexibility to the circuit tuning.

Except at constant added water into the grinding circuit, the improvement of the classification stage leads to increased dilution at the flotation feed and as a consequence to a lower mean residence time, a factor which may offset the benefits of improving the grinding circuit operation. A proper flotation circuit operation must be restored, either by dewatering prior to flotation or by increasing the flotation volume in the rougher-scavenger circuit.

At constant production, the above modifications improve the metallurgical performances, assessed by the net smelter return per ton of ore, as well as the overall net revenue. Since the ore is extremely rich in silver, an increase of capacity, despite a decrease of the net smelter return per ton of ore, can lead to an important increase of the net revenue. The potential improvement is limited by the actual equipment capacity of the grinding and flotation circuits, or simply by the mine output. A method is proposed for maximising the net revenue while avoiding too a large decrease of the metallurgical performances, by maintaining the amount of particles finer than 74 μm in the grinding circuit product. This keeps the silver losses in the high-grade circuit reject at a reasonable level, thus avoiding possible silver losses in the low-grade circuit and a spillage of the ore resource, or a tail retreatment in a far future.

To summarise, the proposed simulation method shows directions for optimising the grinding/flotation operation, which might produce increase of net revenue as large as 20%. Only the high-grade flotation circuit has been considered but it is obvious that there is a potential additional improvement by studying the combined optimisation of the complementary operations of the low and high-grade circuits.

ACKNOWLEDGEMENTS

The authors wish to acknowledge the Peñoles group personnel for their support and permission to publish this paper. Special thanks are offered to Mr. Guadalupe Velazquez from Fresnillo mill and Mrs. Diana Carian Calderon of the C.I.D.T., for the technical support in the laboratory tests and the sampling campaigns. Also the Quebec and Canada Natural Resources Departments, as well as a consortium of fourteen Canadian mining companies, are thanked for their support of the generic research in the field of simulation and control of metallurgical processes.

NOMENCLATURE

AP: hydrocyclone apex diameter

a_i : feed grade in metal i

a, b, c : coefficients in the PSD model

C : concentrate flow rate

c_i : concentrate grade

c_0 : concentrate water content

C_F : fixed costs

C_{MI} : milling costs per unit of time

C_{Mn} : mining cost per unit of time
 C_{Ri} : refining cost for metal i
 C_S : concentrate smelting cost per ton of concentrate
 C_T : concentrate transportation cost
 C_V : variable costs
 D : hydrocyclone diameter
 d_{50c} : corrected cut size
 E_q, E_q' : plant metallurgical efficiencies normalised with respect to concentrate or feed
 E_{qq} : plant economic efficiency
 E_1 : silver plant metallurgical efficiency E_q' , normalised with respect to the ideal concentrate
 E_1' : E_1 at constant flotation feed % solid
 E_2 : silver plant metallurgical efficiency E_q' , normalised with respect to nominal operating conditions
 E_3 : silver plant economic efficiency normalised with respect to nominal operating conditions
 F, F_{max} : ore feed rate and maximum allowed feed rate
 f_i : feed grade in metal i
 H : hydrocyclone height
 IN : hydrocyclone inlet diameter
 i, j : metal indices
 m : classification curve sharpness
 MRT : mean residence time
 PSD : particle size distribution
 P_i : penalty associated to species i
 R : net revenue per unit of time
 R_c, R_c^* : net smelter return per ton of concentrate
 R_f : hydrocyclone by-pass coefficient
 R_o, R_o^* : net smelter return per ton of ore
 T : tail flow rate
 v_i : value of metal i per ton of concentrate
 VF : vortex finder diameter
 W_1 : water feed rate to sump box
 W_2 : water feed rate to ball mill
 x_i : particle size
 y : cumulative PSD
 α_i : price of metal i
 ρ_i : metal i recovery
 ρ, ρ^* : solid actual and ideal recoveries

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